Rohit Dulku

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EDUCATION

University of Massachusetts Lowell Lowell, MA
Master of Science in Electrical Engineering. GPA: 4.0. *Fall 2025*
Relevant Coursework: Power Systems, Generative AI, Digital Signal Processing, Linear Systems, Printed Electronics, Quantum Electronics

University of Massachusetts Lowell Lowell, MA
Bachelor of Science in Electrical Engineering. UMass Lowell Honors College, Dean's List, Eta Kappa Nu (HKN). GPA: 3.75. *Fall 2024*
Relevant Coursework: Circuit Theory I-II, Electronics I-II, Signals & Systems, Power Electronics, Linear Feedback, Logic Design, Microprocessor Systems, PCB Design, Electromagnetics, Data Communications

Codepath
Web Development. *Fall 2026*

EXPERIENCE

Electrical Engineering Intern/Co-op May 2024 – Sep 2024
Revvity *Hopkinton, MA*
- Designed and modified 3 to 5 PCBs for liquid handling systems, including the Janus and Fontus machines.
- Collaborated with Mechanical, Manufacturing, Software, and Quality Engineering teams to build and validate a Janus machine.
- Created Bill of Materials, test reports, and fabrication files in cross-functional project workflows.
- Programmed PCBs using MPLAB IPE with .hex binary files and created visual connection diagrams in Visio.
- Troubleshot PCBs, updated the Peripump PCB, created 3D models in SolidWorks, and troubleshot Winprep software.

Circuits II Lab Assistant Aug 2024 – Dec 2024
University of Massachusetts Lowell *Lowell, MA*
- Managed two lab sessions of approximately 25 and 10 students, graded lab reports, and provided guidance on circuit problems.
- Assisted students with lab equipment and supported hands-on learning in a circuits laboratory setting.

AI Trainer & STEM Subject Matter Expert (Freelance) Apr 2026 – Present
Outlier AI *Remote*
- Evaluated and processed large language model (LLM) outputs across STEM subject areas, completing 15 or more long-form tasks.
- Generated training data using Supervised Fine-Tuning (SFT) to improve model responses and reasoning.

AI Trainer (Contract) Jun 2026 – Jul 2026
Handshake AI *Remote*
- Evaluated and tested Generative AI models specializing in circuit design and simulation.
- Designed complex technical prompts to test LLM design capabilities.
- Executed and reviewed peer-designed workflows as an evaluator to validate circuit outputs.
- Used KiCad, LTspice, NGspice, and SparkPCB to verify schematics and simulations.
- Analyzed model outputs for accuracy, clarity, and depth across circuit design subfields.

Crew Member Nov 2020 – Present
Dunkin' Donuts *Somerville, MA*
- Provided customer service, processed transactions, and maintained a clean and efficient work environment.

PROJECTS

VSCode Web Development Fall 2026

Capstone Project: Lighted Wall Panel Fall 2024
- Designed a 48-cell lighted wall panel for adults over 21 to improve arm and hand strength, with Arduino managing audio, brightness, and knob input via a custom PCB designed in Altium.

Digital Measuring Cup Spring 2024
- Designed an Arduino-based liquid level indicator using water level sensors and an LCD display to measure volume and depth, with LED and buzzer alerts for threshold levels.

Butterworth Filter Design Fall 2023
- Designed and analyzed a Butterworth filter, determining the cutoff frequency using MATLAB; built and tested the circuit using Multisim and an Oscilloscope.

Electromagnetics Stub Matching Fall 2023
- Simulated double stub matching using Smith Charts and CST Studio to analyze impedance matching solutions.

Traffic Light Arduino Fall 2021
- Programmed a two-direction traffic light system using Arduino and breadboard, with red, yellow, and green LEDs cycling in opposing sequences to simulate real intersection logic.

SKILLS

PCB & Hardware Design: Altium 365, KiCad, LTspice, Multisim, Qucs-S, Ngspice, MPLAB IPE, Quartus Prime, Diptrace
Hardware & Equipment: Oscilloscope, Function Generator, Breadboarding, Soldering, 3D Printing, High Voltage Machines
Software & Tools: MATLAB, Arduino, SolidWorks, Visio, Raspberry Pi, Matplotlib, C++, Oracle Agile PLM, Microsoft Office Suite, Adobe
Languages: English (Fluent), Punjabi (Fluent), Hindi (Fluent)

CERTIFICATIONS

- Claude Code 101 Certification (Jul 2026)
- IEEE - Eta Kappa Nu Certification (Nov 2024)
- Real Estate Salesperson's License (Aug 2023)

AWARDS & HONORS

- UMass Lowell Dean's List & Scholarships (2021 - Present)
- Eta Kappa Nu Award (Fall 2024)
- Somerville High School Community Scholarship (2021)

ACTIVITIES & LEADERSHIP

Connexion Nov 2019 - Mar 2020
- Assisted with public meal distributions in Somerville, MA.

East Somerville Main Streets Fall 2019
- Volunteered at local community events.

Let's Get Ready Summer 2019
- Participated in college prep mentorship program.

Robotics, Math & Science League, Community Service
- Active member in school-based STEM and community clubs.